

Seokha Moon

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About Me

I am a Ph.D Student at Korea University, [Vision & AI lab](#) (Advisor: Prof. Jinkyu Kim). Before joining the Vision & AI Lab, I earned a Bachelor's degree in Computer Science from Yonsei University. My research interests lie in the field of Computer Vision, particularly **autonomous robots and autonomous driving**. Recently, I have been focusing on camera-based perception tasks in autonomous driving, including 3D detection and occupancy prediction. I also explored trajectory prediction with an emphasis on modeling interactions between agents, using vision-driven text guidance as supervision to inform the model of relevant contextual cues needed to understand each agent's situation. Currently, I am particularly interested in Vision-Language Navigation for autonomous robots and End-to-End frameworks for autonomous driving.

Education

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|----------------|--|------------------------|
| MS/Ph.D | Korea University , Computer Science and Engineering | Sep. 2022 to Present |
| | <ul style="list-style-type: none"> GPA: 4.36/4.5 Advisor: Prof. Jinkyu Kim | |
| BS | Yonsei University , Computer Science | Mar. 2016 to Aug. 2022 |
| | <ul style="list-style-type: none"> GPA: Overall 3.45/4.3 Major 3.97/4.5 Coursework: Operating System, Computer Architecture, Algorithms, Data Structures, Computer Graphics, Object-Oriented Programming, Artificial Intelligence, Computational Theory. | |

Experience

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| NAVER LABS , Robot & AD Localization, Research Intern | Gyeonggi-do, South Korea
Sep. 2024 to Mar. 2025
6 months |
| <ul style="list-style-type: none"> Developed a efficient yet effective 3D occupancy prediction model (DisOcc). Achieved SOTA performance while maintaining real-time inference speed and reducing memory usage by more than half compared to previous multi-frame fusion approaches. | |
| 42dot. , Motion prediction engineer, Intern | Seoul, South Korea
Aug. 2022 to Dec. 2022
4 months |
| <ul style="list-style-type: none"> Developed algorithms to infer correlations between agents moving at intersections. Researched a teacher-student model incorporating a student model that efficiently considers interactions between agents by mimicking teacher networks and teacher models that jointly consider the future movements of other agents. | |
| Korea Univ. Vision & AI Lab Internship , Intern | Seoul, South Korea
Jan. 2022 to Aug. 2022
9 months |
| <ul style="list-style-type: none"> Implemented state of the art Multi-view 3D Object Detection model (ORA3D) Developed algorithms for the occupancy prediction challenge using the Waymo Open Dataset. | |
| Samsung Electronics , Face Detection Project, Intern | Gyeonggi-do, South Korea
Mar. 2021 to June 2021
4 months |
| <ul style="list-style-type: none"> Implemented few shot face detection algorithm using attention mechanism. Created face dataset and designed a pipeline to augment dataset. | |

Publications

- Image-Guided Semantic Pseudo-LiDAR Point Generation for 3D Object Detection** 2026
Minseung Lee, [Seokha Moon](#), Seung Joon Lee, Reza Mahjourian, Jinkyu Kim
We introduce ImagePG, an image-guided pseudo-LiDAR point generation method that improves 3D object detection for sparse and distant objects while reducing false positives.
Winter Conference on Applications of Computer Vision (WACV)
- SUPER-AD: Semantic Uncertainty-aware Planning for End-to-End Robust Autonomous Driving** 2025
Wonjeong Ryu, Seungjun Yu, [Seokha Moon](#), Hojun Choi, Junsung Park, Jinkyu Kim, Hyun-jung Shim
SUPER-AD introduces uncertainty-aware BEV planning with lane-following priors for safer and more robust camera-only end-to-end autonomous driving.
Preprint
- Stream and Query-guided Feature Aggregation for Efficient and Effective 3D Occupancy Prediction** 2025
[Seokha Moon](#), Janghyun Baek, Giseop Kim, Jinkyu Kim, Sunwook Choi
We propose DuOcc, an efficient 3D occupancy prediction model that maintains dense voxel representations via dual aggregation with stream-based accumulation and query-guided refinement.
Work done during an internship at NAVER LABS.
Preprint
- Who Should Have Been Focused: Transferring Attention-based Knowledge from Future Observations for Trajectory Prediction** 2024
[Seokha Moon](#), Kyuhwan Yeon, Hayoung Kim, Seong-Gyun Jeong, and Jinkyu Kim
We propose an attention-based teacher-student method that distills future-aware interaction knowledge to improve multi-agent trajectory prediction, yielding safer and more human-like predictions.
International Conference on Pattern Recognition (ICPR)
- VisionTrap: Vision-augmented Trajectory Prediction Guided by Textual Descriptions** 2024
[Seokha Moon](#), Hyun Woo, Hongbeen Park, Haeji Jung, Reza Mahjourian, Hyung-gun Chi, Hyerin Lim, Sangpil Kim and Jinkyu Kim
We introduce a novel approach incorporating visual features from surround-view cameras and textual descriptions for trajectory prediction.
European Conference on Computer Vision (ECCV)
- Learning Temporal Cues by Predicting Objects Move for Multi-camera 3D Object Detection** 2024
[Seokha Moon](#), Hongbeen Park, Jaekoo Lee, and Jinkyu Kim
We propose a Detection After Prediction (DAP) method to explicitly learn the temporal cues.
IEEE International Conference on Robotics and Automation (ICRA)
- BEVMap: Map-Aware BEV Modeling for 3D Perception** 2024
Mincheol Chang, [Seokha Moon](#), Reza Mahjourian and Jinkyu Kim
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
- Enhancing Trajectory Prediction Accuracy with Goal Location and Lane Information** 2023
[Seokha Moon](#) and Jinkyu Kim
We introduce a method to predict the goal location and interact with lane to navigate towards the predicted goal.
Conference on The Institute of Electronics and Information Engineers (IEIE)
Oral

RUFI: Reducing Uncertainty in behavior prediction with Future Information

2023

Seokha Moon, Sejeong Lee, Hyun Woo, Kyuhwan Yeon, Hayoung Kim, Seong-Gyun Jeong, and Jinkyu Kim

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop on Vision-Centric Autonomous Driving

ORA3D: Overlap Region Aware Multi-view 3D Object Detection

2022

Wonseok Roh, Gysam Chang, Seokha Moon, Giljoo Nam, Chanyoung Kim, Younghyun Kim, Jinkyu Kim and Sangpil Kim

British Machine Vision Conference (BMVC)

Teaching

Mathmatics for Computer Science (COSE111)

Spring 2024

Algorithm (COSE214)

Fall 2023

Algorithm (COSE214)

Fall 2022

Honors And Awards

First Prize of Computer Science, Yonsei University

Aug. 2021

HUAWEI scholarship, Talent Development Foundation

Nov. 2020

Skills

Languages: C++, C, Python, JavaScript

FRAMEWORKS & TOOLS: PyTorch, OpenCV, TensorFlow, Git, Visual Studio, XCode